Probability Arcade Mini Project
Student Briefing Handout
Random Variables, PMFs, Simulation, R Visualisation, and Interactive Game Design

Materials created under GE Fund OSALP – for authorised educational use only.
Programme: Nurturing Future Risk Managers and Actuaries: An AI-powered Journey for Gifted Secondary Students

1 Mission Brief

Mission Brief

In this mini project, your group will choose one everyday situation involving randomness and turn it into a probability investigation.
Your group will design a small HTML Canvas app or visual simulation, collect data from repeated trials, and use R to produce suitable tables or graphs. The final project should help your audience understand an uncertain situation more clearly.
This project is not only about coding. It is also not only about calculation. It is about using probability, simulation, data visualisation, and mathematical reasoning together.

Key Principle

The central question of the project is:

How can probability help us understand, predict, or make better decisions in this uncertain situation

A strong project should have a clear random mechanism, at least one meaningful random variable, a faithful simulation, useful R output, and a thoughtful conclusion.

2 Main Learning Goals

Through this project, you should practise the following skills:

- describing a random experiment clearly;
- defining at least one random variable;
- explaining how the values of a random variable are generated;
- estimating probabilities from repeated simulation;
- constructing or interpreting an empirical probability distribution;
- recognising discrete probability structures from real situations;
- using R to produce tables or graphs from simulation results;
- using HTML Canvas to build a visual or interactive simulation;
- comparing mathematical reasoning with computational evidence;
- communicating a clear and thoughtful conclusion.

3 What Your Group Should Produce

Project Requirement

Your group should produce three connected parts:

Part A: A probability investigation. Your group should explain the situation, what is random, what quantity you decide to measure, and what your investigation reveals.

Part B: An HTML Canvas app or visual simulation. Your app should make the random process visible or interactive. It may be simple, but it should be meaningful.

Part C: R tables or graphs. Your R output should help answer your probability question. It should support the investigation rather than serve as decoration.

Reminder

You may use AI-assisted coding, but your group must understand the main logic of the app, the simulation, and the analysis. You should be able to explain your work in your own words.

4 Exact Solution and Simulation

Key Principle

Your group should investigate whether the problem can be solved exactly at your current level.

You are not required to force an exact mathematical solution. Instead, you should think carefully about the relationship between mathematical reasoning and simulation.

Some investigations may lead to a simple exact argument. Some may lead to a partial argument. Some may be too difficult to solve exactly at this stage. If an exact solution is difficult, simulation becomes an important tool for exploring the pattern.

Reflection

A weak project only says, "The computer gives this answer."

A stronger project explains what was simulated, what was measured, what pattern was observed, and why the result is reasonable.

5 Presentation Format

Your group will present the project next week.

Slides are not mandatory. You may use any format that helps your group communicate clearly. Your presentation may involve a live app demonstration, a webpage, R output, a short spoken explanation, a poster-style summary, slides, or another suitable format.

The format is flexible. The explanation should still be organised, mathematically meaningful, and understandable to the audience.

6 What Your Presentation Should Communicate

By the end of your presentation, the audience should understand:

1. what situation your group investigated;
2. what is random in the situation;
3. what random variable your group defined;
4. how your simulation works;
5. what your R output shows;
6. whether an exact solution seems accessible at your current level;
7. what your group discovered.

You do not need to follow this list as a fixed presentation structure. These ideas should come through naturally in your explanation.

7 Important Modelling Principle

Key Principle

Although the project topics are written as everyday stories, each one has a precise probability structure. Your group may design the interface, characters, visuals, and presentation style freely, but you should preserve the main random mechanism of your chosen topic.
If you change the rules of the situation, explain clearly how your change affects the probability model.

8 Project Menu

Each group should choose one topic from the following ten topics.

Topic 1: Michael's Pocket Battery Crisis

Michael carries two identical power banks, one in each pocket. At the start, both power banks have the same number of battery bars.

Whenever Michael needs to charge his phone, he randomly chooses one of the two pockets. If the chosen power bank still has battery, one battery bar is used. If the chosen power bank is already empty, Michael only discovers this at that moment.

Your group should investigate what usually remains in the other power bank when Michael first discovers that the chosen power bank is empty.

The essential probability structure is that two equal starting supplies are used randomly, one unit at a time, and the process stops when an already-empty supply is selected.

Topic 2: Ally's Online Toy Collection Challenge

Ally wants to complete a set of mystery keychains. There are several different designs, and every pack contains exactly one randomly chosen design. Each design is equally likely in each pack, and the result of one pack does not affect the result of the next pack.

Ally keeps opening packs until she has collected every design at least once.

Your group should investigate how many packs Ally may need before completing the whole collection.

The essential probability structure is that each draw gives one independent random item from the same set of equally likely item types, and the aim is to collect all types.

Topic 3: Anthony's Recess Coin Duel

Anthony and Michael play a token duel. Anthony starts with some tokens, and Michael starts with some tokens. In each round, a random result determines who wins the round.

If Anthony wins the round, Michael gives one token to Anthony. If Michael wins the round, Anthony gives one token to Michael. The duel continues until one player has no tokens left.

Your group should investigate how starting tokens and round-winning probability affect the final winner and the length of the game.

The essential probability structure is that two players have finite resources, one unit is transferred each round, and the process stops when one player has no resources left.

Topic 4: Keith's Talent Show Audition

Keith is choosing one performer for a school talent show. A fixed number of students audition one by one in random order.

After each audition, Keith must either choose that performer immediately or reject the performer forever. He cannot return to someone he has already rejected, and he cannot see future performers before making the current decision.

Your group should investigate what kind of decision strategy gives Keith a good chance of selecting the strongest performer overall.

The essential probability structure is that choices are made sequentially, candidates appear in random order, and each accept-or-reject decision is irreversible.

Topic 5: The Shared Birthday Notification

Ally is looking at a student group chat. Each student has a birthday, and for this model, assume each birthday is equally likely to be any one of the 365 days in a year.

Your group should investigate how likely it is that at least two students in the group share the same birthday, and how this chance changes as the group size increases.

The essential probability structure is that each person independently receives one of 365 equally likely possible birthdays, and the event of interest is at least one match.

Topic 6: Michael vs Ally: The Mind-Reading Coin Battle

Michael and Ally play a repeated strategy game. In each round, both secretly choose one of two possible options at the same time. Neither player sees the other player's choice before deciding.

If the two choices match, Michael wins the round. If the two choices are different, Ally wins the round.

Your group should investigate whether predictable behaviour can be exploited and whether deliberate randomness can become a useful strategy.

The essential probability structure is simultaneous two-option choices, payoff determined by match versus mismatch, and repeated play under different strategies.

Topic 7: Ally's Extreme Lucky Coin App

Ally finds a mobile game based on repeated coin tosses. The game continues until a target result first appears. The longer Ally waits for that first target result, the larger the reward becomes. In particular, the reward doubles each time the target result is delayed by one more toss.
Your group should investigate the reward pattern of this game and whether the game is as attractive as it first seems.
The essential probability structure is that coin tossing continues until the first success, and the payoff doubles according to the waiting time.

Topic 8: Anthony's Two-Coin Mystery Box

Anthony prepares three mystery boxes for a school booth. Each box contains exactly two tokens.
One box contains two gold tokens. One box contains two silver tokens. One box contains one gold token and one silver token.
A player randomly chooses one box. Then one token from that chosen box is revealed. If the revealed token is gold, the player must judge what this tells them about the other token in the same box.
Your group should investigate how seeing one revealed token changes the probability of what remains hidden.
The essential probability structure is that there are three two-token boxes with fixed compositions, a random box selection, a token reveal, and a conditional question after observing a gold token.

Topic 9: School Leadership Vote Live Count

Ally and Keith are running for a student leadership role. Ally receives more total votes than Keith. However, the votes are mixed together and revealed one by one in a random order during a live count.

Your group should investigate whether Ally's final victory means she is likely to lead throughout the entire count, or whether the order of revealed votes can tell a more complicated story.
The essential probability structure is that one candidate has more total votes, the vote order is random, and the event of interest is whether the leading candidate stays ahead at every stage of the count.

Topic 10: Lost-and-Found Hoodie Mix-Up

After a school activity, several students leave their hoodies in the classroom. Each hoodie belongs to exactly one student. Later, the hoodies are randomly returned to the students without checking the names.
Your group should investigate how many students are likely to receive their own hoodie by chance.
The essential probability structure is that n labelled objects are randomly matched to n labelled people, and the quantity of interest is the number of correct matches.

9 Assessment Rubric

Criterion	What We Are Looking For
Faithful probability modelling	Your group preserves a clear random mechanism in the chosen scenario and explains how the app, simulation, and analysis match that mechanism.
Random variable and distribution thinking	Your group defines at least one meaningful random variable and connects the investigation to discrete probability ideas from the tutorial.
Simulation	Your simulation represents the situation correctly and produces useful evidence for your investigation.
R analysis	Your R tables or graphs help explain the probability pattern and support your conclusion.
Canvas app or visualisation	Your app or visual simulation helps the audience see or interact with the random process.
Exact solution versus simulation	Your group discusses whether an exact solution seems accessible at your current level and explains the role of simulation.
Insight	Your group presents a thoughtful, surprising, or useful conclusion.
Communication	Your final presentation is clear, organised, and engaging.

10 Flexible Final Checklist

Reminder

Before presenting, your group should check that your work can answer the following questions:

- What situation did we investigate?
- What is random in the situation?
- What random variable did we define?

- How does our simulation follow the rules of the situation?
- What does our R output show?
- What have we learned about exact reasoning and simulation?
- What is our main conclusion?

Copyright, Compliance, and Attribution

Key Principle